Jhon Henry Trujillo Montenegro, Ph.D.

jhtrujillomonte@gmail.com
https://www.researchgate.net/profile/Jhon-Trujillo
https://orcid.org/0000-0001-9336-584X
https://scholar.google.com/citations?user=lmq0jmcAAAAJ&hl
https://www.linkedin.com/in/jhtrujillomonte/
https://github.com/jhtrujillo

Summary

Computer Science Engineer and Bioinformatics Scientist with over 12 years of experience in biotechnology and computational approaches for genetic improvement. Specialized in complex polyploid crops, particularly sugarcane, with strong expertise in genomic data analysis and integration. Extensive experience in genetic characterization, complex genome assembly, and next-generation sequencing (NGS) data analysis. Proven expertise in comparative and functional genomics, metagenomics, transcriptomics, genetic mapping, and integration of genomic and phenotypic data for breeding applications. Experienced in working closely with plant breeders, biotechnologists, and multidisciplinary teams, contributing to the design and execution of scientific research projects. Demonstrated leadership in the coordination and management of scientific initiatives, bridging computational and biological domains to support decision-making in breeding programs. Proficient in Unix/Linux environments and programming languages including Python, R, Java, PHP, and Bash. Strong background in algorithm development, bioinformatics pipelines, and software solutions. Demonstrated ability to manage multiple projects, meet deadlines, and work effectively both independently and in collaborative environments.

Education

2012 – 2020 **Ph.D in Computer Science, Universidad del Valle, Cali-Colombia**.
Thesis title: *Genome assembly of a hybrid sugarcane (Saccharum spp.) and construction of molecular fingerprint using High Performance Sequencing. Meritorial doctoral thesis.*
International Internship: *Earlham Institute. Norwich, England*
Full Scholarship by Colciencias-Colfuturo

2002 – 2009 **B.S. Computer Science, Universidad Javeriana, Cali-Colombia**
Thesis title: *Design of a 2D Coordinate Capture Library of Human Movements*.

Professional Experience

2016 – Present **Colombian Sugarcane Research Center, CENICAÑA:** Bioinformatics Area Leader
Development of bioinformatic tools to select the best varieties of sugarcane, including the first assembly of a Colombian sugarcane hybrid and the construction of a molecular fingerprint using SNP markers and Next-Generation Sequencing data. I currently support projects related to genomic selection, metagenomics, assembly of viruses and pathogen genomes, genome-wide association (GWAS), and genetic diversity studies, covering mapping and variants detection with the goal of improving molecular selection projects.

2015 – 2016 **Unión Temporal Gebix:** Research Assistant in Bioinformatics. Cali, Colombia
Improvement of a Java library using Next Generation Sequencing data of human samples to calculate the fractal dimension necessary to identify breast cancer.

Professional Experience (continued)

2008 – 2015 **International Park Of Creativity:** Research Assistant. Manizales, Colombia. *Establishment of computational models for biological systems using Neural Networks to predict the presence of petroleum oil in the field.*

2014 – 2014 **Axure Technologies S.A:** Consultant in Science and Technology. Development of an optical sensor to detect petroleum oil. Bogotá, Colombia. *Construction of predictive models using neuronal networks and designing optical devices for the identification of petroleum oil in the field*

Research Publications

Journal Articles

1. Jaimes, H., Londoño, A., Saavedra-Diaz, C., Trujillo-Montenegro, J. H., López-Gerena, J., Riascos, J. J., & Aguilar, F. S. (n.d.). Sequencing vs. amplification for the estimation of allele dosages in sugarcane (saccharum spp.) *Applications in Plant Sciences*, *n/a*(n/a), e11574. doi:https://doi.org/10.1002/aps3.11574. eprint: https://bsapubs.onlinelibrary.wiley.com/doi/pdf/10.1002/aps3.11574

2. Garcia-Abadillo, J., Adunola, P., Aguilar, F. S., Trujillo-Montenegro, J. H., Riascos, J. J., Persa, R., … Jarquín, D. (2024). Sparse testing designs for optimizing predictive ability in sugarcane populations. *bioRxiv*. doi:10.1101/2024.03.14.584687. eprint: https://www.biorxiv.org/content/early/2024/03/16/2024.03.14.584687.full.pdf

3. Riccio-Rengifo, C., Ramirez-Castrillon, M., Sosa, C. C., Aguilar, F. S., Trujillo-Montenegro, J. H., Riascos, J. J., … Rocha, C. (2024). Assessing drought stress in sugarcane with gene expression and phenomic data using csi-oc. *Industrial Crops and Products*, *216*, 118621.

4. Saavedra-Díaz, C., Trujillo-Montenegro, J. H., Jaimes, H. A., Londoño, A., Villareal, F. A. S., López, L. O., … Quevedo, Y. M. et al. (2024). Genetic association analysis in sugarcane (saccharum spp.) for sucrose accumulation in humid environments in colombia. *BMC Plant Biology*, *24*(1), 570.

5. Trujillo-Montenegro, J. H., Rodriguez Cubillos, M. J., Loaiza, C. D., Quintero, M., Espitia-Navarro, H. F., Salazar Villareal, F. A., … Riascos, J. J. (2021). Unraveling the genome of a high yielding colombian sugarcane hybrid. *Front Plant Sci*, *12*, 694859. doi:10.3389/fpls.2021.694859

6. Zhang, J., Zhang, X., Tang, H., Zhang, Q., Hua, X., Ma, X., … Bowers, J. et al. (2018). Allele-defined genome of the autopolyploid sugarcane saccharum spontaneum l. *Nature genetics*, *50*(11), 1565–1573. doi:https://doi.org/10.1038/s41588-018-0237-2

Pending Articles

1. Trujillo-Montenegro, F. S. A. (2025). *A closer look to the genetic base of sugarcane (Saccharum spp).* In writing process.

2. Trujillo-Montenegro, S. F. (2025). *Optimal whole genome sequencing for* SNP*s detection*. In writing process.

Books and Chapters

1. Trujillo Montenegro, J. H., Martinez Villa, M. C., & Riascos Arcos, J. J. (2024). *Biotecnología para el mejoramiento de la caña de azúcar*. Cali, Colombia: Centro de Investigación de la Caña de Azúcar de Colombia (Cenicaña). Retrieved from https://www.cenicana.org/wp-content/uploads/2024/02/CA-Biotecnologia-para-el-mejoramiento-de-la-cana-de-azucar-31-01-2024-FLIPBOOK.htm

Patents

1. Rengifo, R. C., Trujillo, J. H., & Castillo, J. M. R. (2016, March). Methods and systems for quantifying the grade of petroleum oil based on fluorescence. US Patent App. 14/892,367. Google Patents.
2. Rengifo, R. C., Trujillo, J. H., Castillo, J. M. R., & Palacios, M. C. (2016, April). Compositions and methods for removing petroleum oil from soil. US Patent App. 14/892,226. Google Patents.
3. Rengifo, R. C., Trujillo, J. H., Castillo, J. M. R., & Cubillos, N. Q. (2014, October). Methods and systems for detecting and quantifying petroleum oil based on fluorescence. US Patent App. 14/359,138. Google Patents.

International Presentations

1. Saavedra-Díaz, C. S., Jaimes, H. A., & Trujillo-Montenegro, J. H. (2023). Integración de herramientas de genómica para entender la productividad en caña de azúcar (saccharum spp.) para características de interés [paper presentation]. XII Congreso Tecnicaña. Asociación Colombiana de Técnicos de la Caña, 2022. Cali-Colombia.
2. Trujillo-Montenegro, J. H., Rodriguez Cubillos, M. J., Loaiza, C. D., Quintero, M., Espitia-Navarro, H. F., Salazar Villareal, F. A., … Duitama, J. et al. (2023a). A consensus monoploid genome assembly of the sugarcane variety cc 01-1940, a high yielding colombian sugarcane hybrid in the colombian industry [paper presentation]. ISSCT Molecular Biology Section Chairperson, XXXI Congress, Hyderabad, India.
3. Trujillo-Montenegro, J. H., Rodriguez Cubillos, M. J., Loaiza, C. D., Quintero, M., Espitia-Navarro, H. F., Salazar Villareal, F. A., … Duitama, J. et al. (2023b). Ensamblaje del genoma de la variedad cc 01-1940, lo que hemos aprendido de su contenido genético [paper presentation]. XII Congreso Tecnicaña. Asociación Colombiana de Técnicos de la Caña, 2022. Cali-Colombia.
4. Trujillo, J. H., Duitama, J., Loaiza, C. D., Quintero, M., De Vega, J., & Riascos, J. (2018). Unraveling the genome of a high yielding colombian sugarcane hybrid. In *Plant and animal genome xxvi conference (january 13-17, 2018)*. PAG, San Diego, CA, USA.

Directed Theses

2026

- **Master's Thesis: Genome assembly, structural and functional annotation of orange rust (Puccinia kuehnii) and brown rust (Puccinia melanocephala), phytopathogens of sugarcane**
  Student: Angie Valeria Quiñones Olaya
  Institution: Universidad Del Valle, Colombia.
- **Master's Thesis: Analysis of microbial and functional diversity present in sugarcane juices from three chopping qualities of mechanized harvesting**
  Student: Juanita Sierra Becerra
  Institution: Universidad Del Valle, Colombia.

Honors and Awards

2012-2

- **Recognition given by Organization of American States (OAS/OEA)**, Contribution of technological and economic development through research for developing countries. San Antonio, Tx, USA.

Honors and Awards (continued)

2012-1 **Invited by the Chamber of Commerce in the United States** Junior scientist participating in the development of the patent: Methods and systems for detecting and quantifying petroleum oil based on Fluorescence. Washington DC, USA.

2010 **Recognition given by the Senate of the United States for invention in petroleum oil sensor** Recognition of Scholastic Achievement for the United State Senate – Richard G. Indianapolis, Indiana, USA.

2007,2010 **Recognition given by Massachusetts Institute of Technology (MIT)** Scientific competitions in Synthetic Biology and Computational Modeling. Cambridge massachusetts, USA.

Skills

Coding Java, php, C++, Python, R, Bash, sql, xml/xsl, LaTeX, Nextflow, Docker.

Databases Mysql, Postgresql, sqlite.

Web Dev Html, css, JavaScript, Apache Web Server, Tomcat Web Server.

English Strong reading, writing and speaking competencies

Spanish Native

Referees

John J. Riascos, PhD **Main director of breeding program of the Research Center sugar cane Colombia, CENICAÑA.**
mail: *jjriascos@cenicana.org*

José De Vega, PhD **Main research at Earlham Institute, Norwich Research Park, Norwich, United Kingdom**
mail: *jose.devega@earlham.ac.uk*

Jorge Duitama, PhD **Assistant teacher, Systems and Computing Engineering Department, Universidad de los Andes. Bogotá, Colombia**
mail: *ja.duitama@uniandes.edu.co*